

Introduction to Discrete Probability: Solve the following problems

8. In the city of Milford, applications for zoning changes go through a two-step process: a review by the planning commission and a final decision by the city council. At step 1 the planning commission reviews the zoning change request and makes a positive or negative recommendation concerning the change. At step 2 the city council reviews the planning commission's recommendation and then votes to approve or to disapprove the zoning change. Suppose the developer of an apartment complex submits an application for a zoning change. Consider the application process as an experiment.
 - a. How many sample points are there for this experiment? List the sample points.
 - b. Construct a tree diagram for the experiment.
9. Simple random sampling uses a sample of size n from a population of size N to obtain data that can be used to make inferences about the characteristics of a population. Suppose that, from a population of 50 bank accounts, we want to take a random sample of four accounts in order to learn about the population. How many different random samples of four accounts are possible?
10. Many students accumulate debt by the time they graduate from college. Shown in the following table is the percentage of graduates with debt and the average amount of debt for these graduates at four universities and four liberal arts colleges (*U.S. News and World Report, America's Best Colleges*, 2008).

University	% with Debt	Amount(\$)	College	% with Debt	Amount(\$)
Pace	72	32,980	Wartburg	83	28,758
Iowa State	69	32,130	Morehouse	94	27,000
Massachusetts	55	11,227	Wellesley	55	10,206
SUNY—Albany	64	11,856	Wofford	49	11,012

- a. If you randomly choose a graduate of Morehouse College, what is the probability that this individual graduated with debt?
- b. If you randomly choose one of these eight institutions for a follow-up study on student loans, what is the probability that you will choose an institution with more than 60% of its graduates having debt?
- c. If you randomly choose one of these eight institutions for a follow-up study on student loans, what is the probability that you will choose an institution whose graduates with debts have an average debt of more than \$30,000?
- d. What is the probability that a graduate of Pace University does not have debt?
- e. For graduates of Pace University with debt, the average amount of debt is \$32,980. Considering all graduates from Pace University, what is the average debt per graduate?

11. The National Highway Traffic Safety Administration (NHTSA) conducted a survey to learn about how drivers throughout the United States are using seat belts (Associated Press, August 25, 2003). Sample data consistent with the NHTSA survey are as follows.

Region	Driver Using Seat Belt?	
	Yes	No
Northeast	148	52
Midwest	162	54
South	296	74
West	252	48
Total	858	228

- For the United States, what is the probability that a driver is using a seat belt?
 - The seat belt usage probability for a U.S. driver a year earlier was .75. NHTSA chief Dr. Jeffrey Runge had hoped for a .78 probability in 2003. Would he have been pleased with the 2003 survey results?
 - What is the probability of seat belt usage by region of the country? What region has the highest seat belt usage?
 - What proportion of the drivers in the sample came from each region of the country? What region had the most drivers selected? What region had the second most drivers selected?
 - Assuming the total number of drivers in each region is the same, do you see any reason why the probability estimate in part (a) might be too high? Explain.
12. The Powerball lottery is played twice each week in 28 states, the Virgin Islands, and the District of Columbia. To play Powerball a participant must purchase a ticket and then select five numbers from the digits 1 through 55 and a Powerball number from the digits 1 through 42. To determine the winning numbers for each game, lottery officials draw five white balls out of a drum with 55 white balls, and one red ball out of a drum with 42 red balls. To win the jackpot, a participant's numbers must match the numbers on the five white balls in any order and the number on the red Powerball. Eight coworkers at the ConAgra Foods plant in Lincoln, Nebraska, claimed the record \$365 million jackpot on February 18, 2006, by matching the numbers 15-17-43-44-49 and the Powerball number 29. A variety of other cash prizes are awarded each time the game is played. For instance, a prize of \$200,000 is paid if the participant's five numbers match the numbers on the five white balls (Powerball website, March 19, 2006).
- Compute the number of ways the first five numbers can be selected.
 - What is the probability of winning a prize of \$200,000 by matching the numbers on the five white balls?
 - What is the probability of winning the Powerball jackpot?

18. To investigate how often families eat at home, Harris Interactive surveyed 496 adults living with children under the age of 18 (*USA Today*, January 3, 2007). The survey results are shown in the following table.

Number of Family Meals per Week	Number of Survey Responses
0	11
1	11
2	30
3	36
4	36
5	119
6	114
7 or more	139

For a randomly selected family with children under the age of 18, compute the following.

- The probability the family eats no meals at home during the week.
 - The probability the family eats at least four meals at home during the week.
 - The probability the family eats two or fewer meals at home during the week.
19. The National Sporting Goods Association conducted a survey of persons 7 years of age or older about participation in sports activities (*Statistical Abstract of the United States*, 2002). The total population in this age group was reported at 248.5 million, with 120.9 million male and 127.6 million female. The number of participants for the top five sports activities appears here.

Activity	Participants (millions)	
	Male	Female
Bicycle riding	22.2	21.0
Camping	25.6	24.3
Exercise walking	28.7	57.7
Exercising with equipment	20.4	24.4
Swimming	26.4	34.4

- For a randomly selected female, estimate the probability of participation in each of the sports activities.
- For a randomly selected male, estimate the probability of participation in each of the sports activities.
- For a randomly selected person, what is the probability the person participates in exercise walking?
- Suppose you just happen to see an exercise walker going by. What is the probability the walker is a woman? What is the probability the walker is a man?

20. *Fortune* magazine publishes an annual list of the 500 largest companies in the United States. The following data show the five states with the largest number of *Fortune* 500 companies (*The New York Times Almanac*, 2006).

State	Number of Companies
New York	54
California	52
Texas	48
Illinois	33
Ohio	30

Suppose a *Fortune* 500 company is chosen for a follow-up questionnaire. What are the probabilities of the following events?

- Let N be the event the company is headquartered in New York. Find $P(N)$.
 - Let T be the event the company is headquartered in Texas. Find $P(T)$.
 - Let B be the event the company is headquartered in one of these five states. Find $P(B)$.
21. The U.S. adult population by age is as follows (*The World Almanac*, 2009). The data are in millions of people.

Age	Number
18 to 24	29.8
25 to 34	40.0
35 to 44	43.4
45 to 54	43.9
55 to 64	32.7
65 and over	37.8

Assume that a person will be randomly chosen from this population.

- What is the probability the person is 18 to 24 years old?
- What is the probability the person is 18 to 34 years old?
- What is the probability the person is 45 or older?